

Name of Student :	Pramod Jadhav
Semester / Section :	8th - B
Registration No :	22070706
Roll No :	CSE - 145
Internship Offered By :	IT Education Nagpur
Duration of Internship :	1th January - 30th May 2026
Stipend Amount Received Per month :	Upto 5000



Office of Dean (Training & Placement Cell)

Ref. : YCCE/T&P/2026-25/CSE/

Date: 16-01-2026

To

Mr. Rajesh Pawar,
HR, IT Education
Nagpur

Subject: - Regarding internship at your esteemed organization.

Yeshwantrao Chavan College of Engineering (YCCE) having permanent AICTE ID **1-4736951** established in 1984, is the flagship Engineering Institute of **Meghe Group of Institutions (MGI)**, a well-known name in the field of Education in Central India creating Mind Share and Human Capital to cater to the requirements of nation building.

In the recently declared "**National Institute Ranking Framework**" (NIRF) by the Ministry of HRD Government of India for the top Institutes under type **A** category which included IITs, NITs, Government Colleges, Private and Autonomous colleges YCCE is ranked in TOP 150.

YCCE have been accredited by NAAC with Grade "A++".

As our endeavor to continuously upgrade the technical skill of our students we are very much thankful to you for allowing our students (as per the communication with department of Computer Science and Engineering) to undergo **four months Internship** in your esteemed organization as per the agreed norms, for a period as mentioned below:

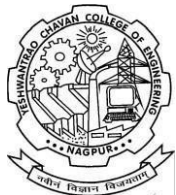
Sr. No.	Name of Intern Students	Semester/Year	Engineering Branch
1	Mr. Pramod Jadhav	8th Semester / 4th Year	B.Tech – Computer Science and Engineering

Duration of Internship – From 01/01/2026 to 31/05/2026

Kindly allow the candidate to give presentations based on internship work, as it is a part of evaluation. You will have to share the month wise feedback of student. Hope your positive response towards same. Thanking you.

Dr. Lalit B. Damahe
HoD (Dept. of Computer Science
and Engineering)
Email: hod_cse@ycce.edu

Dr. Rakhi D. Wajgi
Dean (Training & Placement)
Mobile : 9970238062
Email : dean_tp@ycce.edu



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Science and Engineering

Date: 16/01/2026

Undertaking by Student

This **Semester Long Internship** is a part of my B. E. VIIIth Semester degree course curriculum under which I will be evaluated. I have completely read the attached guidelines of Semester Long Internship. During the entire period, from Date 01/01/2026 to date 31/05/2026 of **Semester Long Internship**, I, **Pramod Jadhav** shall strictly abide by the rules and regulations, safety guidelines, of **IT Education Nagpur**, failing which my Internship shall be terminated with immediate effect. In addition to this, I will follow the COVID-19 norms, from the date of reporting at industry. Yeshwantrao Chavan College of Engineering Nagpur or **IT Education Nagpur** will not be responsible in case of mishap/eventualities happen.

I am aware that the Evaluation Report is to be submitted to Faculty Mentor / Guide with the consent of Industry persons/Trainers. This evaluation report along with TWO Internship Presentations carries total 100 marks which will be clubbed with ER of 2 credits in my SoE of 8th semester.

Name & Signature of Student
Pramod Jadhav –

M. No: 8719831717

Undertaking by Parent

I, being the Parent of **Pramod Jadhav**, do hereby undertake that my ward is going on **Semester Long Internship** to **IT Education Nagpur** as a part his curriculum. I undertake that my son **Pramod Jadhav** studying in B.E. VIIIth Semester shall follow the rules and guidelines of college as well as Industry.

Name & Signature of Parent
Govind Jadhav –

M. No: 8888012186



Nagar Yuwak Shikshan Sanstha's

Form No YI-03

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

(Accredited 'A' Grade by NAAC)

Hingna Road, Wanadongri, Nagpur - 441 110

Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu

E-mail : principal@ycce.edu, info@ycce.edu

Department of Computer Science and Engineering

ATTENDANCE SHEET

Name & Address of Organization

IT Education

Nagpur, India

Name of Student	Pramod Jadhav	
Roll. No	CSE - 145	
Name of Course	Computer Science and Engineering	
Date of Commencement of Internship/Training:	01/01/2026	
Date of Completion of Internship/Training:	30/05/2026	

Month & Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Jan 2026										H	H			H			H	H						H	H	H					H
Feb 2026	H						H	H					H	H						H	H						H				
March 2026	H			H			H	H					H	H						H	H						H	H			
April 2026				H	H						H	H						H	H						H	H					

Rajesh

Signature of Company internship supervisor

Name - Rajesh Pawar

Contact No. 9665302573



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E-mail : principal@ycce.edu, info@ycce.edu

Department of Computer Science and Engineering

STUDENT'S DIARY/ LOG

Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 5 January 2026 To : 16 January 2026

Week: 1,2	DATE : 05/01/2026	16/01/2026	
Dept./Division	Computer Science and Engineering	Name of Project	Portfolio Application
Name of HOD/ Supervisor With e-mail id	Rajesh Pawar hr@iteducations.in		
Main points of the week	Developed a Personal Portfolio Website		
The Personal Portfolio Website is a web development project that showcases my technical skills, projects, and achievements in a professional format. In this project, the website is designed and developed using HTML, CSS, and JavaScript, ensuring a fully responsive and user-friendly interface across different devices. It includes sections such as About Me, Skills, Projects, and Contact, allowing easy navigation through a fixed navigation bar and smooth scrolling effects. Basic animations like typing effects and hover interactions are also implemented to enhance user experience. This project demonstrates how front-end technologies can be used to create a visually appealing digital resume that can be shared with recruiters and professionals to highlight my profile and capabilities.			

Rajesh

Signature of Industry Supervisor

Signature of Student

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****STUDENT'S DIARY/ LOG**

Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 19 January 2026 To : 30 January 2026

Week: 3,4		DATE : 19/01/2026	30/01/2026
Dept./Division	Computer Science and Engineering	Name of Project	Console base CRUD operation
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Implemented Console-Based CRUD Operations in Java	
The Console-Based CRUD Operation project is a Java application that manages data through a command-line interface. In this project, users interact with the system by entering inputs in the console to perform operations such as creating, reading, updating, and deleting records. Core Java concepts like classes, objects, loops, and conditional statements are used to handle program logic and user interaction. The application allows users to add new records, view existing data, modify details, and delete unwanted entries efficiently. Data can be stored using arrays, ArrayLists, or basic file handling depending on the implementation. This project demonstrates how fundamental programming concepts can be applied to build a simple data management system and provides a strong foundation for understanding CRUD operations in software development.			

Signature of Industry Supervisor

Signature of Student

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****STUDENT'S DIARY/ LOG**

Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 2 February 2026 To : 13 February 2026

Week: 5,6		DATE : 02/02/2026	13/02/2026
Dept./Division	Computer Science and Engineering	Name of Project	Http request
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Implemented Form Handling using GET, POST, and DELETE Requests (Frontend to Backend)	
The Form Handling project focuses on managing user input by sending data from the frontend to the backend using different HTTP methods such as GET, POST, and DELETE. In this project, forms are created using HTML and styled with CSS, while JavaScript is used to handle user interactions and send requests to the server. The GET method is used to retrieve data, POST is used to submit and store user input, and DELETE is implemented to remove data from the database. The backend processes these requests, performs the required operations, and sends appropriate responses back to the frontend. This project demonstrates how client-server communication works in web development and provides practical understanding of handling user data efficiently using RESTful principles.			

Signature of Industry Supervisor

Signature of Student

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Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 16 February 2026 To : 27 February 2026

Week: 7,8		DATE : 16/02/2026	27/02/2026
Dept./Division	Computer Science and Engineering	Name of Project	Student Management System
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Developed a Student Management System using Java Swing	

The Student Management System is a desktop-based application designed to digitize and simplify the process of managing student records. In this project, a graphical user interface is developed using Java Swing, providing an easy-to-use and structured layout with forms, buttons, and tables for efficient interaction. The system allows users to add student details such as name, roll number, and course, view records in a tabular format, update existing data, delete unwanted entries, and search for specific records quickly. Database integration is achieved using JDBC with SQLite/MySQL to ensure secure storage and retrieval of data. The application follows CRUD operations, where users can create, read, update, and delete student records seamlessly. This project demonstrates how Java-based desktop applications can be used to build efficient data management systems with real-world usability.

Signature of Industry Supervisor

Signature of Student

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Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 2 March 2026 To : 13 March 2026

Week: 9,10		DATE : 02/03/2026	13/03/2026
Dept./Division	Computer Science and Engineering	Name of Project	Python Frame Work
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Learned and implemented basic Python framework Flask with Jinja templating	
The Flask and Jinja project focuses on building a basic web application using the Flask framework in Python and Jinja as the templating engine. In this project, routes are created in Flask to handle different URLs and user requests, while Jinja is used to dynamically render HTML pages by passing data from the backend to the frontend. The application demonstrates how templates, variables, loops, and conditional statements can be used within HTML to create dynamic content. Basic features such as form handling, page navigation, and data display are implemented to understand the interaction between frontend and backend. This project demonstrates how lightweight Python frameworks can be used to develop simple and scalable web applications efficiently.			

Signature of Industry Supervisor

Signature of Student

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****STUDENT'S DIARY/ LOG**

Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 16 March 2026 To : 27 March 2026

Week: 11,12		DATE : 16/03/2026	27/03/2026
Dept./Division	Computer Science and Engineering	Name of Project	SQL
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Learned and implemented Flask with SQLAlchemy for database management	
The Flask with SQLAlchemy project focuses on integrating a Python web framework with an Object Relational Mapping (ORM) tool to manage databases efficiently. In this project, Flask is used to create the web application and define routes, while SQLAlchemy is used to handle database operations such as creating tables, inserting data, updating records, and deleting entries without writing raw SQL queries. Models are defined in Python classes, and these are automatically mapped to database tables. The application demonstrates how user input from forms can be stored and retrieved from the database dynamically. This project helps in understanding how backend frameworks and ORM tools work together to build scalable and maintainable web applications.			

Signature of Industry Supervisor

Signature of Student

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****STUDENT'S DIARY/ LOG**

Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 30 March 2026 To : 10 April 2026

Week: 13,14		DATE : 30/03/2026	10/04/2026
Dept./Division	Computer Science and Engineering	Name of Project	Tudo using Flask
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Developed a To-Do List Application using Flask and Jinja	
The To-Do List application is a simple web-based project built using the Flask framework in Python and Jinja templating engine. In this project, users can add, view, update, and delete tasks through a web interface. Flask is used to handle routing and backend logic, while Jinja is used to dynamically render HTML pages and display task data. The application allows users to input tasks, mark them as completed, and manage their daily activities efficiently. Data is processed on the backend and displayed in real-time on the frontend, ensuring smooth interaction between both layers. This project demonstrates the practical implementation of CRUD operations in a web application and strengthens the understanding of Flask, Jinja, and basic backend development concepts.			

Signature of Industry Supervisor

Signature of Student



Nagar Yuwak Shikshan Sanstha's

Form No YI-04

Yeshwantrao Chavan College of Engineering

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu

E-mail : principal@ycce.edu, info@ycce.edu

Department of Computer Science and Engineering

STUDENT'S DIARY/ LOG

Your Name:

Pramod Jadhav

Your Institute: YCCE, Nagpur

Company where internships are carried out: IT Education

Period : 2 weeks

From : 13 April 2026 To : 24 April 2026

Week: 15,16		DATE : 13/04/2026	24/04/2026
Dept./Division	Computer Science and Engineering	Name of Project	API integration
Name of HOD/ Supervisor With e-mail id		Rajesh Pawar hr@iteducations.in	
Main points of the week		Weather Forecasting Application	
The Weather Forecasting Application project was developed using external APIs to fetch real-time weather data. Users can search for any location and view details such as temperature, humidity, and weather conditions. The application processes API responses and displays them in a clean and interactive UI. Error handling was implemented for invalid inputs and API failures. This project improved understanding of API integration, asynchronous operations, and dynamic data handling.			

Rajesh

Signature of Industry Supervisor

Signature of Student

**YeshwantraoChavan College of Engineering***(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)***(Accredited 'A' Grade by NAAC)**

Hingna Road, Wanadongri, Nagpur - 441 110

Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****STUDENT FEEDBACK OF INTERNSHIP****Student Name:** Pramod Jadhav**Work Supervisor:** Rajesh Pawar**Supervisor Email:** hr@iteducations.in**Internship is:** paid**Company/Organization:** IT Educations, Nagpur**Internship Address:** Hingna Road, Near Lokmanya Metro Station, Nagpur - 440016**Faculty Coordinator:** Prof. Snehal Dongre**Department:** Computer Science
and Engineering**Was your internship experience related to your major area of study?**Yes, to a large degree

Yes, to a slight degree

No, not related at all

This experience has:	Strongly Agree	Agree	No Opinion	Disagree	Strongly Disagree
Given me the opportunity to explore a career field		☒			
Allowed me to apply classroom theory to practice		☒			
Helped me develop my decision-making and problem-solving skills		☒			
Expanded my knowledge about the work world prior to permanent employment		☒			
Helped me develop my written and oral communication skills		☒			
Provided a chance to use leadership skills (influence others, develop ideas with others, stimulate decision-making and action)		☒			
This experience has:	Strongly Agree	Agree	No Opinion	Disagree	Strongly Disagree
Expanded my sensitivity to the ethical implications of the work		☒			

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involved		✓			
Made it possible for me to be more confident in new situations		✓			
Given me a chance to improve my interpersonal skills		✓			
Helped me learn to handle responsibility and use my time wisely		✓			
Helped me discover new aspects of myself that I didn't know existed before		✓			
Helped me develop new interests and abilities		✓			
Helped me clarify my career goals		✓			
Provided me with contacts which may lead to future employment		✓			
Allowed me to acquire information and/ or use equipment not available at my Institute		✓			

- In the Institute internship program, faculty members are expected to be mentors for students. Do you feel that your faculty coordinator served such a function?

Yes

- In what areas did you most develop and improve?

Machine Learning

- What did you dislike about the internship?

Nothing

- Considering your overall experience, how would you rate this internship? (Circle one).
(Satisfactory/ Good/ **Excellent**)

- Give suggestions as to how your internship experience could have been improved.

It would be good if internship were in-person.

**Yeshwantrao Chavan College of Engineering***(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)***(Accredited 'A' Grade by NAAC)**

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****PROFORMA FOR EVALUATION OF INTERNSHIP BY INSTITUTE**

Ph. _____ Fax _____ Email _____

Evaluation(I) _____

- | | | |
|--|--|------------------------------------|
| 1. Name of Student | Pramod Jadhav | Mob. No. 9359072163 |
| 2. College Roll No. | <u>CSE - 145</u> | Enrolment No. 22070706 |
| 3. Branch/Semester | <u>CSE /8th</u> | Period <u>5 Months</u>
Training |
| 4. Home Address with contact No. | <u>At post Sonari, Tr. Bhokar, Dist. Nanded, 9359072163</u> | |
| 5. Address of Training Site: | <u>Work from Home</u> | |
| 6. Address of Training Providing Agency: | <u>Hingna Road, Near Lokmanya Metro Station, Nagpur - 440016</u> | |
| 7. Name/Designation of Training In- charge | Rajesh Pawar | |
| 8. Type of Work | <u>Training-Based</u> | |
| 9. Date of Evaluation | <u>26/04/2026</u> | |
| | a. Attendance: (Satisfactory/ Good/ <u>Excellent</u>) | |
| | b. Practical Work: (Satisfactory/ Good/ <u>Excellent</u>) | |
| | c. Faculty's Evaluation: (Satisfactory/ Good/ <u>Excellent</u>) | |
| | d. Evaluation of Industry: (Satisfactory/ Good/ <u>Excellent</u>) | |
| Overall grade: | (Satisfactory/ Good/ <u>Excellent</u>) | |

Signature of Faculty Mentor

Signature of Internship Supervisor (Industry)

**Yeshwantrao Chavan College of Engineering***(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)***(Accredited 'A' Grade by NAAC)**

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Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.eduE-mail : principal@ycce.edu, info@ycce.edu**Department of Computer Science and Engineering****SUPERVISOR EVALUATION OF INTERN****Student Name:** Pramod Jadhav**Work Supervisor:** Rajesh Pawar**Organization:** It Education**Internship Address:** Nagpur, India**Dates of Internship:** **From** - 1 January 2026**To** - 30 May 2026

Please evaluate your intern by indicating the frequency with which you observed the following behaviors:

Parameters	Needs improvement	Satisfactory	Good	Excellent
Behavior				✓
Performs in a dependable manner				✓
Cooperates with co-workers and supervisors				✓
Shows interest in work/ Shows good judgment				✓
Learns quickly				✓
Shows initiative				✓
Produces high quality work				✓
Accepts responsibility/ Accepts criticism				✓
Demonstrates organizational skills				✓
Uses technical knowledge and expertise				✓
Demonstrates creativity/originality				✓
Analyzes problems effectively				✓
Is self-reliant				✓
Communicates well and Writes effectively				✓
Has a professional attitude				✓
Gives a professional appearance				✓
Is punctual and Uses time effectively				✓



Nagar Yuwak Shikshan Sanstha's

Form No YI-07

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E-mail : principal@ycce.edu, info@ycce.edu

Department of Computer Science and Engineering

Overall performance of student intern (circle one):

(Needs improvement/Satisfactory/Good/Excellent)

Additional comments, if any: NIL

Rajesh



Signature of Industry supervisor

INTERNSHIP REPORT

ON

Full Stack Development

Yeshwantrao Chavan College of Engineering

Bachelor of Technology

in

Computer Science & Engineering



Session 2025-26

Submitted by:

Pramod Jadhav

Supervised by:

Prof. Snehal Dongre

Mr. Rajesh Pawar

Nagar YuwakShikshanSanstha's

YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING,

(An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

NAGPUR – 441 110

2025-26

CERTIFICATE OF APPROVAL

Certified that the internship project entitled “**Full stack developerIntern Applications**” has been successfully completed by Mr. Pramod Jadhav during the academic session 2025–26.

Faculty Supervisor

Signature :

Name : Prof. Snehal Dongre

Designation: Professor (Computer Science and Engineering)

Industry Supervisor

Signature :



Name : Mr. Rajesh Pawar

Designation: Full stack developerIntern

ACKNOWLEDGEMENT

I, Mr. Pramod Jadhav would like to convey my gratitude to Yeshwantrao Chavan College of Engineering *Mr. Rajesh Pawar (HR Head) IT Educations Nagpur.* for emphasizing on the Semester Internship Program and giving me the platform to interact with industry professionals.

I would also like to thank *Prof. Snehal Dongre and Dr. Lalit Damahe (Head of Department)* for giving me the opportunity to work on the prestigious Internship.

I extend my warm gratitude and regards to everyone who helped me during my internship

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EXECUTIVE SUMMARY

INTRODUCTION OF REPORT

This report presents a comprehensive overview of the internship undertaken as part of the academic curriculum in Computer Science and Engineering by Pramod Jadhav, a student of the Bachelor of Technology in Computer Science and Engineering (8th Semester) at Yeshwantrao Chavan College of Engineering. The internship was conducted over a period of five months, from January 2026 to May 2026, at IT Educations, Nagpur, where I worked as a Full Stack Developer Intern. This experience played a significant role in bridging the gap between theoretical knowledge and practical application in a professional environment.

The primary aim of this internship was to gain hands-on experience and understand how academic concepts are applied in real-world projects. It provided valuable exposure to web application development and helped strengthen my technical skills. Throughout the internship, I enhanced my coding abilities, analytical thinking, and problem-solving skills, which are essential for a successful career in software development.

In today's rapidly evolving technological landscape, practical knowledge is as important as theoretical understanding. This internship offered insights into how real-world projects are executed in professional environments. I gained an understanding of teamwork, task allocation, and effective deadline management. Additionally, I became familiar with standard workflows followed in software development organizations.

During the internship, I learned the complete process of developing web applications, starting from requirement analysis to design, development, testing, and maintenance. I observed how proper planning and structured development approaches contribute to building efficient and reliable systems. I also gained experience working with various technologies used in full stack development. This included handling both frontend and backend components, which helped me understand how user interfaces interact with server-side logic and databases. I learned how to integrate multiple modules to develop fully functional applications.

Beyond technical skills, the internship contributed significantly to my personal and professional growth. I improved my communication skills through regular interaction with mentors and team members. I also developed essential soft skills such as teamwork, time management, and adaptability. Working in a professional environment instilled a sense of discipline, responsibility, and commitment to meeting deadlines.

Throughout this period, I actively participated in real-time tasks and project development. This hands-on experience increased my confidence in coding, debugging, and solving practical problems. It also

enhanced my ability to approach challenges systematically and develop effective solutions.

This report provides detailed insights into the organization, tasks performed during the internship, tools and technologies used, and the skills gained. It also highlights the challenges faced and the approaches used to overcome them.

In conclusion, the internship was a highly valuable and enriching experience. It not only strengthened my technical capabilities but also enhanced my professional skills. Overall, it has prepared me for future opportunities in the software development field and laid a strong foundation for career growth.

OVERVIEW OF THE INTERNSHIP PROVIDING ORGANIZATION

The internship was undertaken at IT Educations, Nagpur, a well-established training and development organization that focuses on delivering quality education and practical exposure in the field of Information Technology. The organization aims to bridge the gap between academic knowledge and industry requirements by providing hands-on training, real-time project experience, and professional guidance to students and aspiring IT professionals.

IT Educations is committed to enhancing the technical and professional skills of students by offering a wide range of programs in domains such as web development, full stack development, data analytics, and software engineering. The organization believes in a practical approach to learning, where students are encouraged to actively participate in real-time projects rather than relying solely on theoretical concepts. This approach helps learners understand how technologies are applied in real-world scenarios and prepares them for industry challenges.

Located in Nagpur, IT Educations has built a reputation for providing industry-oriented training and internship opportunities. The organization is equipped with experienced trainers and mentors who guide students throughout their learning journey. These professionals bring valuable industry experience and knowledge, which helps interns gain insights into current trends, tools, and technologies used in the IT sector. Regular mentoring sessions and feedback mechanisms ensure continuous improvement and skill enhancement. The core objective of IT Educations is to make students job-ready by focusing on both technical and soft skills. Along with technical training, the organization also emphasizes communication skills, teamwork, time management, and professional ethics. This holistic development approach ensures that students are well-prepared to meet the expectations of the corporate world.

One of the key features of IT Educations is its project-based learning methodology. Interns are assigned tasks and projects that simulate real-world applications, enabling them to work on practical problems and develop effective solutions. This hands-on experience plays a crucial role in building confidence and improving problem-solving abilities. Interns are also exposed to different phases of software development, including requirement analysis, system design, development, testing, and deployment.

The organization follows a structured workflow similar to that used in professional software companies. Interns are required to understand project requirements, plan their work

accordingly, and complete assigned tasks within specified deadlines. This helps in developing a sense of responsibility and discipline among students. Additionally, working in a team environment allows interns to collaborate, share ideas, and learn from each other. IT Educations also ensures that students are familiar with modern tools and technologies used in the industry. Interns get the opportunity to work with programming languages, frameworks, and databases that are widely used in real-world applications. This exposure helps them stay updated with current industry standards and enhances their employability. During the internship, I had the opportunity to work under the guidance of experienced mentors who provided continuous support and valuable feedback. The organization maintained a positive and professional work environment that encouraged learning and innovation. I was able to understand how tasks are assigned, managed, and executed in a structured manner. This experience gave me a clear understanding of how software development processes function in a real-world setting.

Furthermore, IT Educations focuses on continuous assessment and improvement. Interns are regularly evaluated based on their performance, participation, and project work. Constructive feedback is provided to help them identify their strengths and areas of improvement. This process ensures that students keep progressing and refining their skills throughout the internship period.

The organization also plays a significant role in guiding students toward their career goals. By providing exposure to industry practices and expectations, IT Educations helps interns understand the skills required to succeed in the IT field. It encourages students to think critically, work efficiently, and adapt to changing technological environments.

In conclusion, IT Educations, Nagpur, serves as an excellent platform for students to gain practical knowledge and industry exposure. Its focus on hands-on learning, structured development practices, and overall skill enhancement makes it a valuable organization for aspiring IT professionals. The internship experience at this organization not only strengthened my technical knowledge but also contributed significantly to my personal and professional growth, preparing me for future challenges in the software development industry.

ABOUT THE PROJECT

1. Introduction to the Project

During the internship at IT Educations, I worked on multiple projects that helped me understand real-world software development. These projects were designed to strengthen my skills in both frontend and backend development, along with API integration and database management.

The main projects developed during the internship include:

- Portfolio Application
- Student Management System (Java Swing)
- To-Do List Application (Flask)
- Weather API Integration Project

Each project focused on different concepts such as CRUD operations, UI design, backend logic, and external API usage. These projects provided hands-on experience and helped me understand how complete applications are built and deployed.

2. Project 1: Portfolio Website

2.1 Objective

The objective of this project was to design and develop a personal portfolio website to showcase my technical skills, projects, and achievements in a professional manner. The website acts as a digital resume that can be easily shared with recruiters and professionals to highlight my profile and capabilities.

2.2 Technologies Used

- HTML
- CSS
- JavaScript

2.3 Features

- Fully responsive design for mobile and desktop devices
- Attractive and user-friendly interface
- About Me section with personal and academic details
- Skills section highlighting technical expertise (Python, SQL, Excel, Power BI)
- Projects section showcasing key projects
- Contact section with GitHub, LinkedIn, and Email links
- Smooth scrolling and basic animations (typing effect, hover effects)
- Fixed navigation bar for easy access

2.4 Learning Outcomes

- Developed strong understanding of modern UI/UX design principles
- Gained experience in building responsive layouts using CSS Flexbox
- Learned DOM manipulation using JavaScript for interactivity
- Improved skills in structuring a complete web application
- Understood how to present technical work in a professional portfolio format
- Enhanced ability to design clean, user-friendly, and visually appealing interfaces

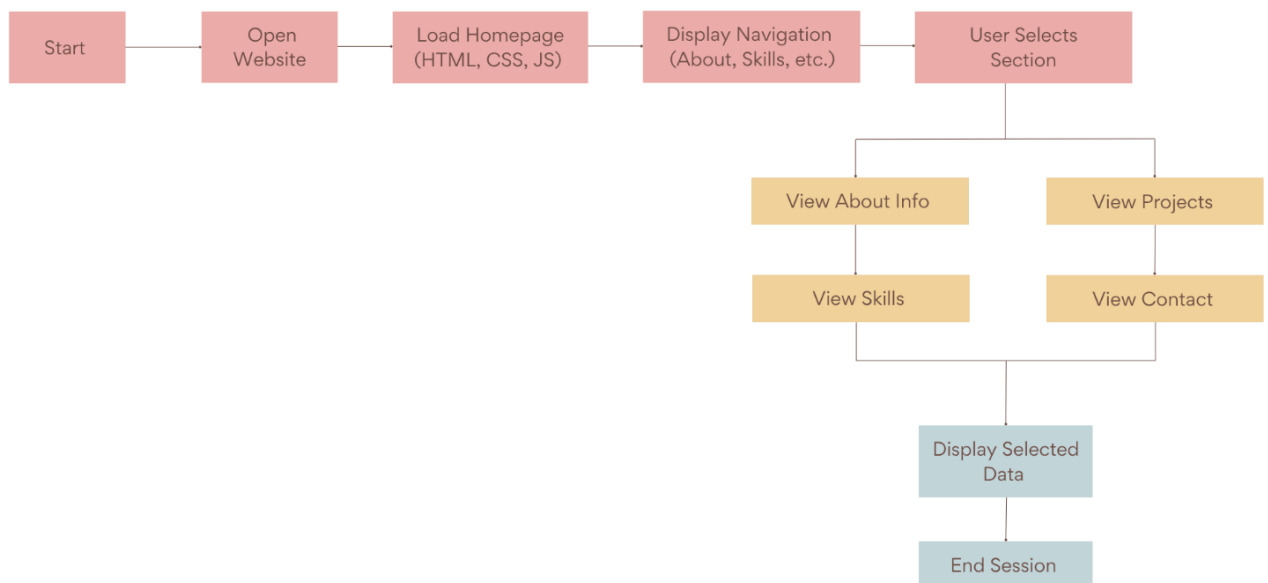


Fig 1: System Flowchart of Portfolio Application

3. Project 2: Student Management System (Java Swing)

3.1 Objective

The objective of this project was to design and develop a desktop-based Student Management System that digitizes the process of managing student records. The system eliminates manual record-keeping by providing an efficient, reliable, and user-friendly interface for handling student data.

3.2 Technologies Used

- Java (Core Programming)
- Java Swing (GUI Development)
- SQLite / MySQL (Database)
- JDBC (Database Connectivity)

3.3 Features (Aligned with UI & Functionality)

- Graphical User Interface (GUI)
- A structured and easy-to-use interface built using Java Swing with forms, buttons, and tables.
- Add Student Module
- Allows users to input student details such as name, roll number, and course.
- View Records (Table View)
- Displays all student data in a tabular format for better readability and management.
- Update Functionality
- Enables modification of existing student records efficiently.
- Delete Functionality
- Allows removal of unwanted or outdated records from the database.
- Search Feature
- Provides quick access to specific student records using search input.
- Database Integration
- Ensures persistent storage and retrieval of data using JDBC connection.

3.4 CRUD Operations (Core Logic)

- Create → Insert new student records into database
- Read → Fetch and display student data
- Update → Modify existing records
- Delete → Remove student entries

3.5 System Workflow

- User enters data through GUI form
- Input is processed using Java event handling
- JDBC establishes connection with database
- Data is stored/retrieved from database
- Output is displayed in table format on UI

3.6 Learning Outcomes

- Strong understanding of Java Swing GUI design
- Implementation of event-driven programming
- Practical experience with database connectivity using JDBC
- Execution of CRUD operations in real-world application
- Basic understanding of application structure and data flow

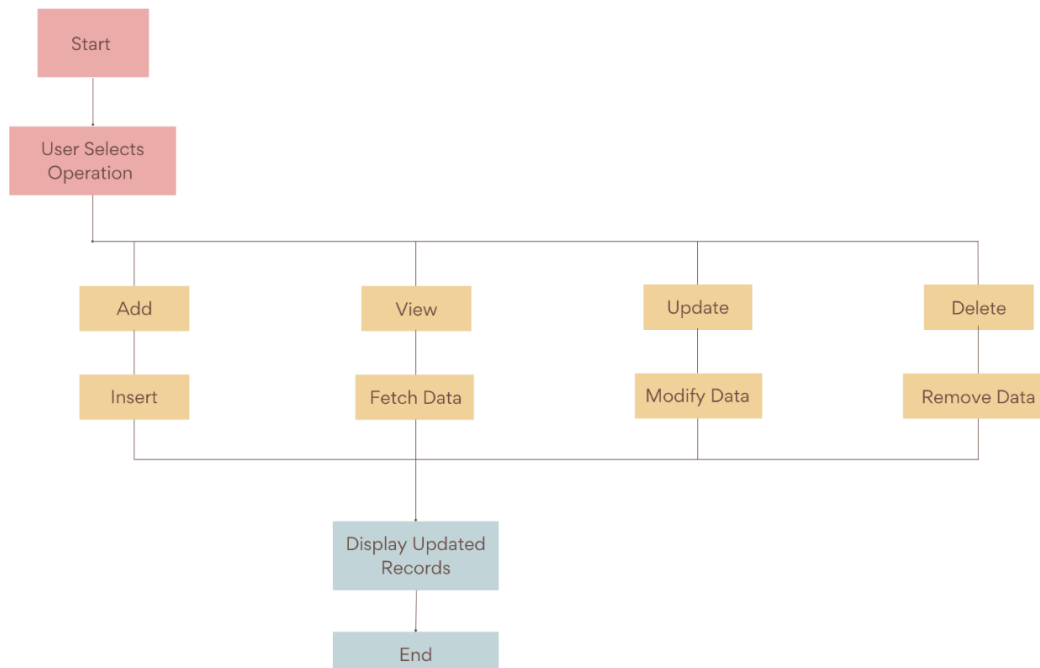


Fig 2: System Flowchart of Student Management System

4. Project 3: To-Do List Application (Flask)

4.1 Objective

The objective of this project was to develop a web-based task management system that allows users to efficiently organize and track their daily activities through a simple and interactive interface.

4.2 Technologies Used

- Python
- Flask (Backend Framework)
- HTML, CSS (Frontend)
- SQLite (Database)

4.3 Features (Aligned with Web App UI)

- **Task Creation Module**
Users can add new tasks through a simple input interface.
- **Task Display Interface**
All tasks are displayed dynamically in a structured list format.
- **Update Task Feature**
Allows modification of existing tasks.
- **Delete Task Option**
Enables removal of completed or unnecessary tasks.
- **Mark as Completed**
Users can track progress by marking tasks as done.
- **Backend Processing with Flask**
Handles routing, request processing, and database interaction.
- **Persistent Data Storage**
Tasks are stored in SQLite database for long-term usage.

4.4 Working (System Behavior)

The application follows a client-server architecture where user actions such as add, update, or delete are sent to the Flask backend. The backend processes the request, interacts with the SQLite database, and returns the updated data to the user interface.

4.5 Learning Outcomes

- Understanding of Flask backend development
- Knowledge of routing and HTTP request handling
- Hands-on experience with database operations (SQLite)
- Implementation of full-stack integration
- Improved understanding of web application workflow

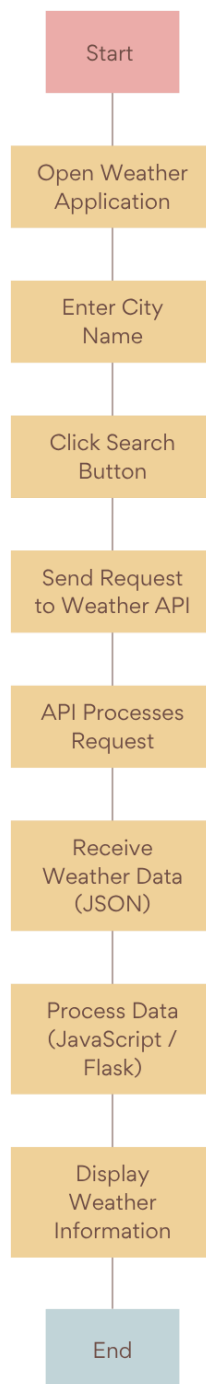


Fig 3: System Flowchart of To-Do List Application using Flask

5. Project 4: Weather Application (API Integration)

5.1 Objective

The objective of this project was to develop a weather application that fetches and displays real-time weather information using an external API based on user input.

5.2 Technologies Used

- Python / JavaScript
- Flask (if backend used)
- OpenWeather API
- HTML, CSS

5.3 Features (Aligned with Functionality)

- **City-Based Search**
Users can enter a city name to get weather details.
- **Real-Time Weather Data Display**
Shows temperature, humidity, and weather conditions.
- **API Integration**
Fetches live data from OpenWeather API.
- **Dynamic Data Rendering**
Displays processed JSON data in a readable format.
- **Error Handling**
Handles invalid city input and API failures gracefully.
- **Simple and Clean UI**
Ensures easy interaction and readability.

5.4 Working (System Behavior)

The user inputs a city name in the interface. The application sends a request to the OpenWeather API. The API returns weather data in JSON format, which is then parsed and displayed dynamically on the frontend. Error handling mechanisms ensure proper response even in case of invalid inputs.

5.5 Learning Outcomes

- Practical knowledge of API integration
- Handling and parsing JSON data
- Implementation of real-time data fetching
- Understanding of client-server communication
- Improved debugging and error handling skills

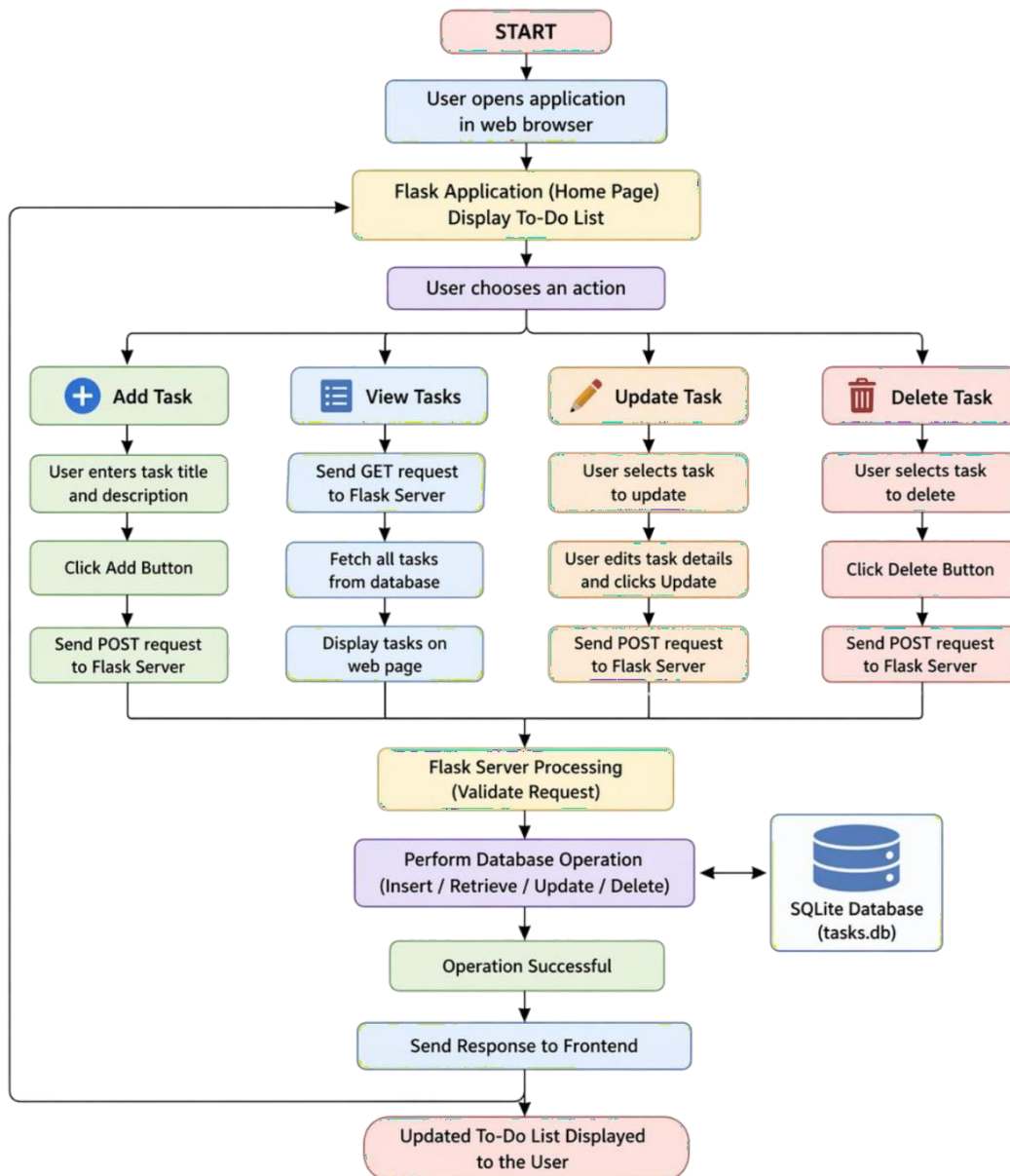


Fig 4: System Flowchart of Weather Application using API

6. Challenges Faced:

- During the development of the projects, several technical and design challenges were encountered:
- Backend debugging issues due to errors in Flask and Java logic
- Database connectivity problems while using JDBC and SQLite
- Difficulty in handling API responses in the Weather Application
- UI responsiveness issues across different screen sizes

7. Solutions:

To overcome the above challenges, the following solutions were implemented:

- Used debugging techniques such as print statements and logs to identify errors
- Implemented proper error and exception handling
- Followed a modular coding approach for better code management
- Adopted a structured development process (design, coding, testing)

8. Skills Gained

- Technical Skills:
- Full Stack Development (Frontend, Backend, Database)
- Java Swing GUI Development
- Flask Backend Development
- API Integration and JSON Handling
- Soft Skills:
- Problem-solving ability
- Time management
- Team collaboration
- Communication skills

9. Need for the Project

In today's digital era, there is a growing need for efficient, automated, and user-friendly software solutions to solve real-world problems. Traditional methods of managing data and tasks are often time-consuming, error-prone, and inefficient. Therefore, the development of modern applications becomes essential to improve productivity, accuracy, and accessibility. The Portfolio Website was developed to create a professional online presence that showcases skills, projects, and achievements, making it easier for recruiters to evaluate a candidate's profile.

The Student Management System was designed to replace manual record-keeping with a digital solution that ensures better organization, quick access, and secure storage of student data.

The To-Do List Application addresses the need for effective task management by helping users organize and track their daily activities efficiently.

The Weather Application fulfills the need for real-time information by providing accurate weather data using API integration.

Overall, these projects demonstrate the importance of software applications in simplifying complex tasks, improving efficiency, and enhancing user experience in various domains.

10. Problem Definition and Analysis

In many real-world scenarios, traditional methods of managing data and daily activities are inefficient, time-consuming, and prone to human errors. There is a need for digital systems that can simplify operations, improve accuracy, and provide quick access to information.

The Portfolio Website project addresses the problem of presenting personal skills and achievements in a structured and professional way. Without a portfolio, it becomes difficult for recruiters to evaluate a candidate's capabilities effectively.

The Student Management System solves the problem of manual record-keeping in educational institutions. Managing student data using paper-based systems can lead to data loss, duplication, and difficulty in retrieval. A digital system ensures secure storage, easy updates, and quick access to records.

The To-Do List Application focuses on improving task management. Many individuals struggle to organize and track their daily tasks efficiently. This application provides a simple platform to manage tasks, improving productivity and time management.

The Weather Application addresses the need for real-time weather information. Users often require accurate and up-to-date weather data for planning daily activities. By integrating an API, the application provides instant weather updates based on user input.

Overall, these projects analyze common real-life problems and provide efficient, user-friendly software solutions that enhance productivity, accuracy, and accessibility.

11. Objectives of the Project

The main objective of this project is to design and develop efficient, user-friendly software applications that solve real-world problems using modern technologies. The project focuses on improving productivity, data management, and user experience through practical implementations.

The specific objectives of the project are as follows:

- To develop a Portfolio Website that professionally showcases skills, projects, and achievements in a structured and visually appealing manner
- To design a Student Management System that digitizes student records and performs CRUD operations efficiently
- To build a To-Do List Application that helps users manage and organize daily tasks effectively
- To create a Weather Application that provides real-time weather information using API integration
- To implement proper frontend and backend integration for smooth data flow between user interface and server
- To use databases for storing and managing application data securely
- To enhance problem-solving and development skills through practical project implementation
- To gain hands-on experience in full stack development, API usage, and software design

12. Methodology

The methodology followed during the internship is based on a structured approach to software development. It includes different phases that ensure systematic planning, development, and successful implementation of projects. The process followed is similar to the Software Development Life Cycle (SDLC).

12.1 Requirement Analysis

In this phase, the requirements of the project are identified and understood. The objectives, functionalities, and expected outcomes of each project are clearly defined. This helps in creating a proper roadmap for development.

12.2 System Design

After understanding the requirements, the system design is prepared. This includes designing the architecture, user interface (UI), database structure, and workflow of the application.

12.3 Development

In this phase, the actual coding is performed. Frontend and backend components are developed using appropriate technologies such as HTML, CSS, JavaScript, Java Swing, and Flask.

12.4 Testing

The developed system is tested to identify and fix errors. Various test cases are applied to ensure that all functionalities are working correctly and efficiently.

12.5 Implementation

After successful testing, the system is deployed and made ready for use. The final application is executed in a real or simulated environment.

12.6 Maintenance

Post-deployment, the system is monitored and updated if required. Any bugs or improvements are handled in this phase.

13. Features of the System

The systems developed during the internship include several important features that ensure efficient performance, user convenience, and reliability. These features are designed to meet user requirements and provide a smooth working experience.

13.1 User-Friendly Interface

All applications are designed with a simple and easy-to-use interface, allowing users to interact with the system without requiring technical expertise.

13.2 CRUD Operations

The system supports all basic operations including Create, Read, Update, and Delete. These operations allow users to efficiently manage data such as student records and task lists.

13.3 Responsive Design

The web-based applications are responsive and can adapt to different screen sizes, including desktops and mobile devices, ensuring better accessibility.

13.4 Database Integration

The applications are connected to databases such as SQLite or MySQL, enabling secure storage and retrieval of data.

13.5 Real-Time Data Handling

The system is capable of handling real-time data, especially in applications like the weather system where live data is fetched using APIs.

13.6 Modular Structure

The applications are developed using a modular approach, where different components are divided into smaller parts, making the system easier to manage and maintain.

13.7 Error Handling

Proper error handling techniques are implemented to manage unexpected situations and prevent system crashes.

13.8 Secure and Reliable

Basic security measures are followed to ensure that user data is handled safely and the system performs reliably.

13.9 Cross-Platform Compatibility

The applications can run on different platforms such as Windows and web browsers without major issues.

13. Outcome

The internship resulted in significant improvement in both technical and professional skills. It provided practical exposure to real-world software development and helped in understanding how theoretical concepts are applied in actual projects. During the internship, multiple applications were successfully developed, which enhanced hands-on experience.

The outcome includes improved knowledge of programming languages and development tools used in building web and desktop applications. It also helped in understanding database management and how data is stored, retrieved, and manipulated efficiently.

Working with different technologies increased adaptability and learning ability.

The internship also strengthened problem-solving and analytical thinking skills. While working on projects, various challenges were encountered, which helped in developing logical approaches to find effective solutions. Debugging and error handling skills were also improved during this process.

In addition, the internship contributed to the development of important soft skills such as communication, teamwork, and time management. Regular interaction with mentors and structured work practices helped in understanding professional work culture.

Overall, the outcome of the internship was highly positive, as it enhanced confidence, improved technical expertise, and prepared me for real-world job opportunities in the field of software development and data analytics.

14. Conclusion of Project

The projects undertaken during the internship provided valuable practical experience in the field of software development and data handling. Through the development of multiple applications such as the portfolio website, student management system, To-Do list application, and weather application, various concepts learned during academic studies were successfully implemented in real-world scenarios.

These projects helped in understanding the complete software development process, including requirement analysis, design, development, testing, and implementation. The use of different technologies such as HTML, CSS, JavaScript, Java Swing, Flask, and databases like SQLite/MySQL enhanced technical knowledge and development skills.

In addition to technical learning, the projects also contributed to improving problem-solving abilities, logical thinking, and debugging skills. Working on these applications provided insight into how real systems are built and maintained in a professional environment.

Overall, the internship and project work have played a significant role in strengthening both technical and professional competencies. It has built a strong foundation for future career opportunities in software development and data analytics.

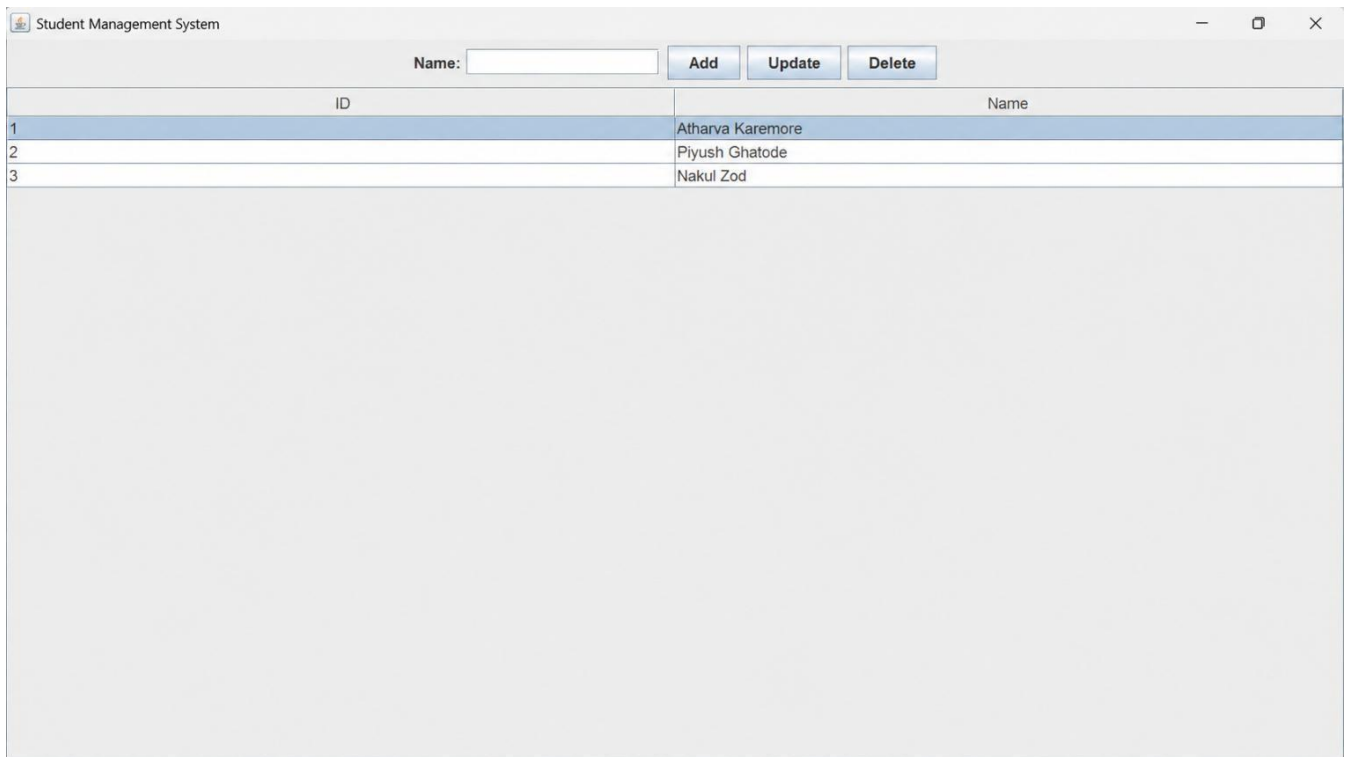


Fig 2: Student Management System (Java Swing) Interface

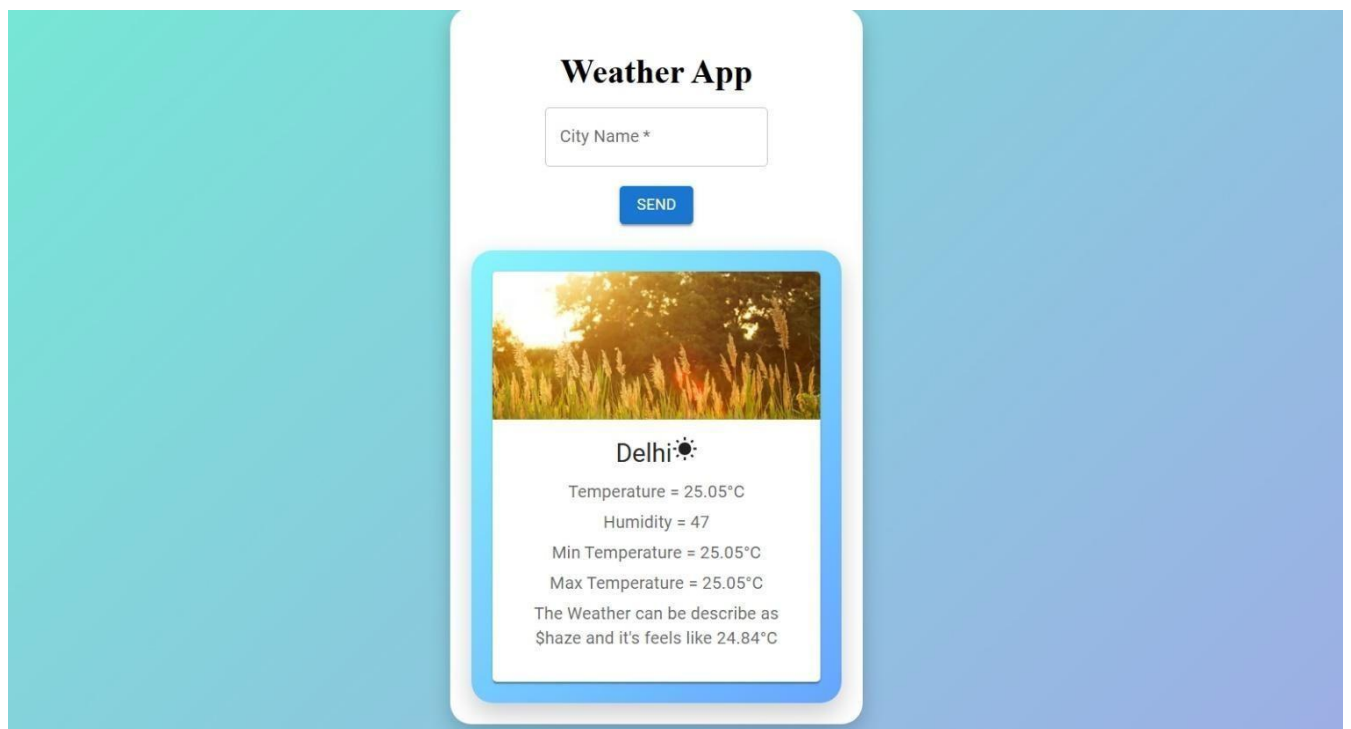


Fig 4: Weather Application (API Integration) Interface

15. DETAIL OVERVIEW OF PROJECT

During the internship, multiple projects were developed to gain practical exposure to real-world application development. Each project focused on different technologies and concepts, helping in building a strong technical foundation.

15.1 Portfolio Website

The Portfolio Website is a personal web application designed to showcase my skills, projects, certifications, and educational background. It acts as a digital resume that can be accessed online. The website is developed using HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface. It includes various sections such as Home, About, Education, Projects, Certificates, and Contact. This project demonstrates my ability to design modern web interfaces and present information in a structured manner.

15.2 Student Management System (Java Swing)

The Student Management System is a desktop-based application developed using Java Swing. It is designed to manage student records efficiently by performing CRUD operations such as adding, viewing, updating, and deleting student data. The system uses a graphical user interface for easy interaction and integrates with a database using JDBC. This project

highlights my understanding of GUI development, event handling, and database connectivity.

15.3 To-Do List Application (Flask)

The To-Do List Application is a web-based task management system developed using the Flask framework. It allows users to organize their daily tasks by adding, updating, deleting, and marking tasks as completed. The application uses SQLite as a database to store task information. It follows a client-server architecture where the frontend interacts with the backend to process user requests. This project demonstrates full stack development concepts and backend logic implementation.

15.4 Weather Application (API Integration)

The Weather Application is designed to provide real-time weather information using an external API. Users can search for weather details of any city, and the system displays temperature, humidity, and weather conditions. The application fetches data from the OpenWeather API and processes it in JSON format. This project demonstrates API integration, real-time data handling, and dynamic content display.

Overall Summary

All the projects together cover a wide range of technologies including web development, desktop application development, backend frameworks, and API integration. These projects helped in understanding different aspects of software development and provided hands-on experience in building functional applications.

CONCLUSION

The internship undertaken as part of the Bachelor of Technology in Computer Science and Engineering program proved to be a highly enriching and transformative experience. It provided a valuable platform to bridge the gap between theoretical academic knowledge and practical industry-oriented skills. Through continuous learning and hands-on involvement, the internship contributed significantly to both technical proficiency and professional development.

During the internship period, multiple projects were designed and developed, including a portfolio website, a student management system, a To-Do list application, and a weather application. These projects covered a wide range of technologies such as frontend development, backend frameworks, database management, and API integration. Working on these applications enabled a deeper understanding of how different components of software systems interact to deliver complete and functional solutions.

The experience also provided exposure to the complete Software Development Life Cycle (SDLC), including requirement analysis, system design, implementation, testing, and maintenance. Understanding these stages helped in realizing the importance of structured development, proper planning, and systematic execution in building efficient and scalable applications.

From a technical perspective, the internship strengthened skills in programming, debugging, database connectivity, and real-time data handling. It also enhanced the ability to analyze problems logically and develop effective solutions. Exposure to concepts such as CRUD operations, RESTful APIs, and modular development approaches contributed to building a strong technical foundation.

In addition to technical growth, the internship played a crucial role in improving soft skills. Regular interaction with mentors and peers helped in developing communication skills, teamwork, adaptability, and time management. Working under deadlines and handling multiple tasks improved the ability to perform efficiently in a professional environment.

Furthermore, the internship increased confidence in working on real-world applications and facing practical challenges. It encouraged a proactive learning approach and developed the ability to explore new technologies independently. This experience has not only enhanced current capabilities but also created a strong base for continuous learning and career growth. In conclusion, the internship was a comprehensive learning experience that successfully

combined academic knowledge with practical implementation. It has prepared me to take on future challenges in the field of software development and data analytics with confidence and competence. The skills and knowledge gained during this period will serve as a strong foundation for professional success and long-term career development.